PHYS 3200 - Fall 2026

End of Topic Quiz 4

October 8th, 2026

Name and ID #:

Instructions:

1. You may use:
   - A calculator.
   - Notes.
   - Writing utensil: pencil and eraser are best.
2. If you have a question about a problem (confused about the situation, need some missing piece of information, etc.), please raise your hand and ask!
3. $\boxed{\text{Box your final answer}}$ for each part.
4. Credit will not be given if your answers are too messy or obscure to read. Remember: a grader who isn't squinting and spending extra time trying to decipher mysterious scratches is a happy grader.
5. This quiz must be turned in by the end of the class period.

1. A particle of mass m moves under the influence of the attractive central force

$$\mathbf{F}(r) = -\frac{k}{r^2}\hat{\mathbf{r}},$$

where k is a positive constant. Take the potential energy to be zero at infinite separation:

$$U(\infty) = 0.$$

(a.) Explain why this force is a central force.

(b.) Using:

$$\mathbf{F}(r) = -\frac{\mathrm{d}U}{\mathrm{d}r}\hat{\mathbf{r}},$$

determine the potential-energy function $U(r)$. Clearly show the integral and apply the condition $U(\infty) = 0$.

(c.) The particle is released from rest at a distance r_0 from the force center. Write the particle's initial total mechanical energy.

(d.) Use conservation of energy to find the particle's radial speed $|\dot{r}|$ when it reaches a smaller distance r, where $r < r_0$.

2. Two particles, each of mass m, move along the x-axis and interact through a spring. The potential energy of the two-particle system is

$$U(r) = \frac{1}{2}k(r-\ell)^2,$$

where $r = x_2 - x_1$ is the distance between the particles, ℓ is the spring's equilibrium length, and k is a positive constant.

Initially, the particles are at rest and separated by a distance

$$r_0 = \ell + d,$$

where $d > 0$. The particles are then released.

(a.) Find the initial total energy of the system.

(b.) Explain why the total momentum of the two-particle system is conserved. What is its value?

(c.) Use momentum conservation to prove $\dot{x}_2 = -\dot{x}_1$.

(d.) Use conservation of energy to find $\dot{x}_1$ and $\dot{x}_2$ when $r = \ell$. Assume particle 1 is initially to the left of particle 2.